



Engineering Sciences

AI for Autonomous Systems

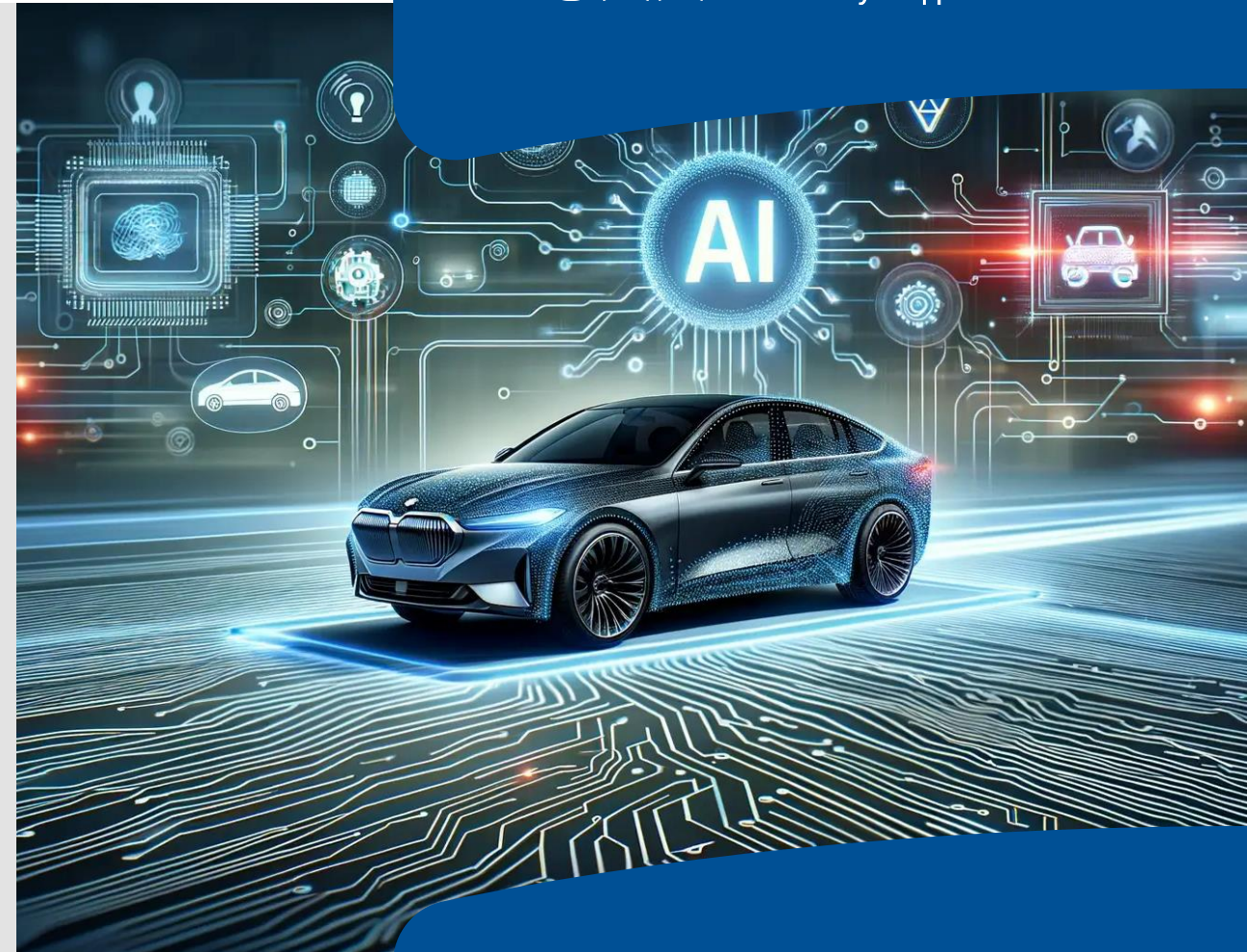
KI Summer School 2026

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Agenda

1. Introduction
2. Sensors
3. AI for Image & Obstacle Recognition
4. Obstacle and Person Detection
5. Sensor Fusion
6. Localization
7. Path and Trajectory Planning
8. State of the Art
9. Our RC Car
10. Conclusion

Introduction



The Key Question

“What capabilities does a system need in order to act sensibly and safely without direct human control?”

| Automation vs. Autonomy

According to the Fraunhofer IESE definition, there are fundamental technical differences:



Automation

Follows fixed, deterministic rules. Its behavior is precisely predictable (e.g., industrial robots on an assembly line).



Autonomy

The system takes responsibility for its own actions and makes independent decisions. It reacts to unforeseen events.



AI-Based

Artificial intelligence serves primarily as a powerful tool (e.g., machine learning) to enable this autonomy in the first place.



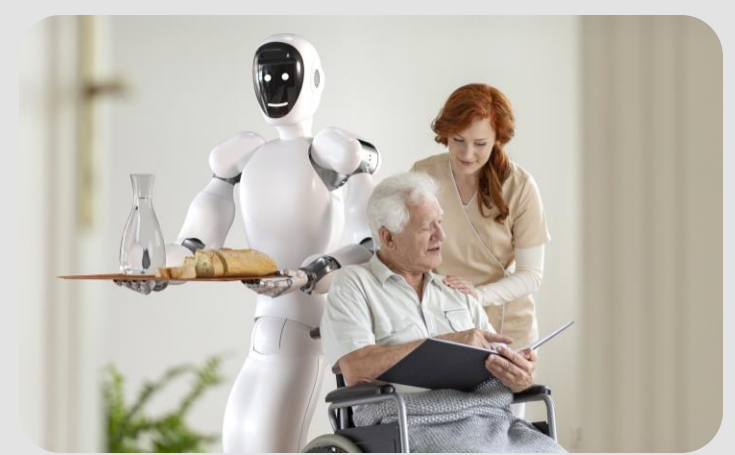
| Anwendungsbereiche in der Praxis



Logistics Robots
in a Modern Warehouse



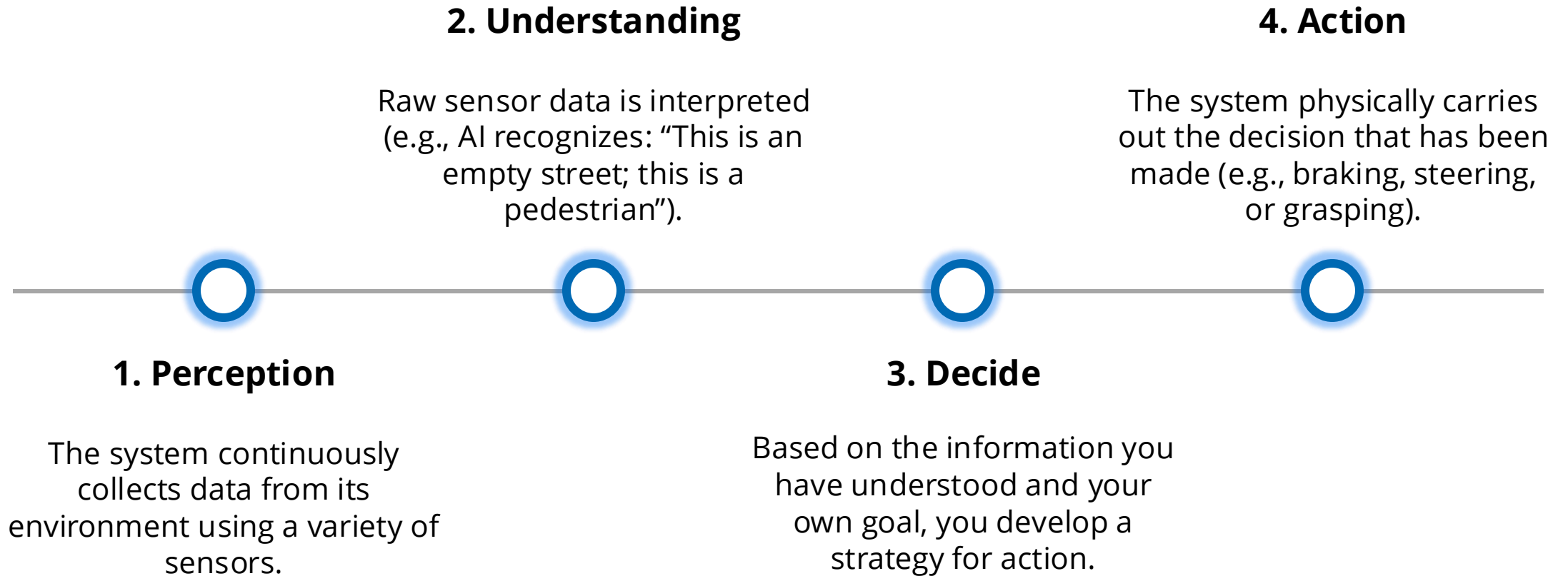
Autonomous Driving
in Road Traffic



Humanoid Robots
in Nursing and Personal Care

| The Sense-Plan-Act Cycle

The universal fundamental principle by which all autonomous systems operate at all times:



| Anatomy of a Secure System

Secure Autonomous System

AI-based pattern
recognition

highly reliable
classical
algorithms

state-of-the-art
sensor technology

mathematical
optimization

robust classical
control theory

Sensors

How does an autonomous system perceive its environment?



How does a system perceive its environment?

A vehicle has no eyes or ears.

Instead, it relies on a highly complex array of sensors that continuously collect data from the environment.

This network of sensors is the autonomous system's only connection to physical reality. Before an AI can make a decision, the world must be measured with precision.



The “Big Three” of Environmental Monitoring



Camera

- ✓ Detects colors and textures (e.g., traffic lights, signs)
- ✓ Very high resolution
- ✗ Ineffective in darkness, fog, or backlighting
- ✗ No direct distance measurement



LiDAR

- ✓ Millimeter-precise 3D point cloud of the surroundings
- ✓ Works absolutely perfectly in the dark
- ✗ Interfered with by heavy rain or snowfall
- ✗ Hardware is expensive and often mechanically complex



Radar

- ✓ Weather is completely irrelevant (penetrates rain and fog)
- ✓ Direct measurement of speeds (Doppler)
- ✗ Very coarse spatial resolution
- ✗ Detects “something,” but has difficulty classifying objects

| Short-Range & Internal Sensors



Ultrasound

Application: Unbeatably affordable for distances in the centimeter range (e.g., parking assist).

Disadvantage: Sound travels too slowly for high-speed driving and is distorted by airflow.



Inertial Measurement Unit (IMU)

Application: The “balance organ.” Continuously measures acceleration and rotational rates. Extremely fast; requires no external signals.

Disadvantage: Measurement errors accumulate over time (drift).



Wheel Encoder

Application: Measures wheel rotation to determine the vehicle's exact forward movement.

Disadvantage: In the event of wheel slip (ice, wet conditions, spinning tires), the measured distance deviates from reality.



GPS / GNSS

Application: Provides the absolute location on a world map for global navigation.

Disadvantage: Often accurate to only a few meters (too imprecise for lane keeping) and loses signal in tunnels.

The Illusion of the Perfect Sensor

Why isn't a single sensor enough?



| The Illusion of the Perfect Sensor

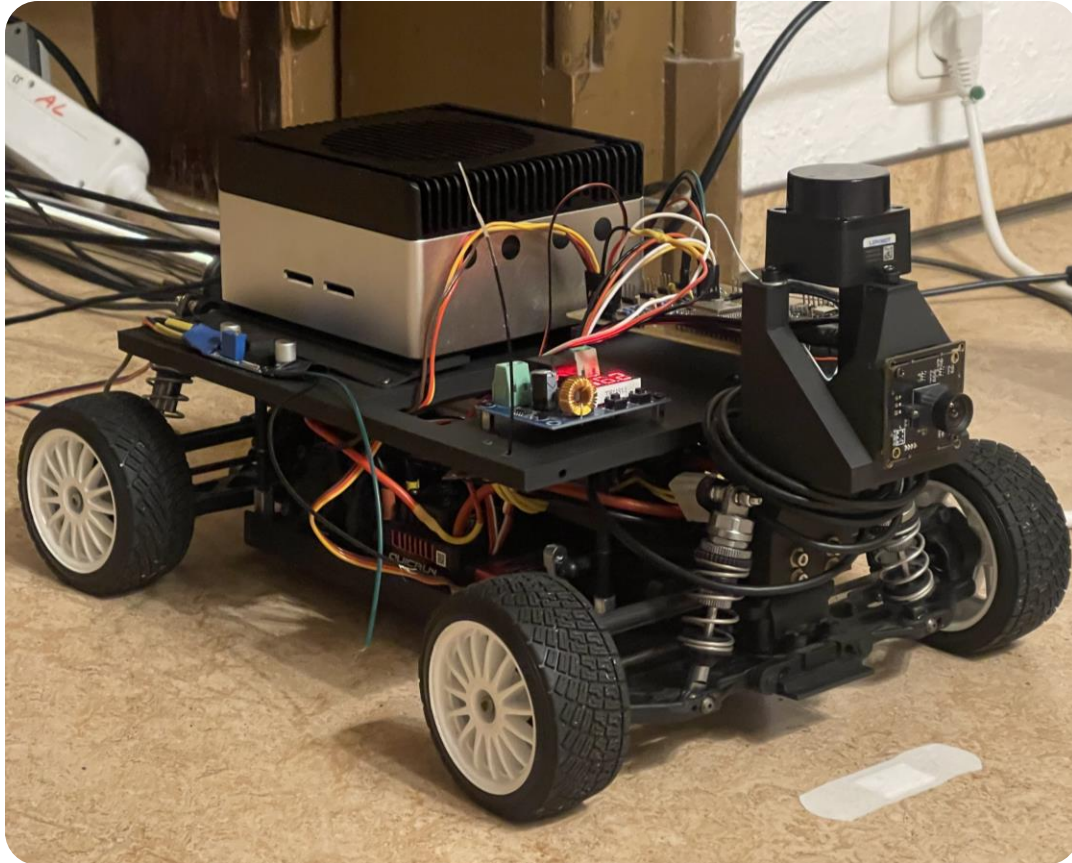
Why isn't a single sensor enough?

Every single sensor has its own specific “blind spot.” If a system relies blindly on a single data source, sudden changes in weather or physical limitations can lead to catastrophic errors.

The solution is sensor fusion: Only by combining and cross-checking multiple sensors (redundancy) can a reliable overall system be created that isn't easily fooled.



| Example Using Our Vehicle



CAMERA

> "There's an orange pylon up ahead."

LIDAR

> "Confirmed. Detecting object exactly 2.00 m away."

IMU

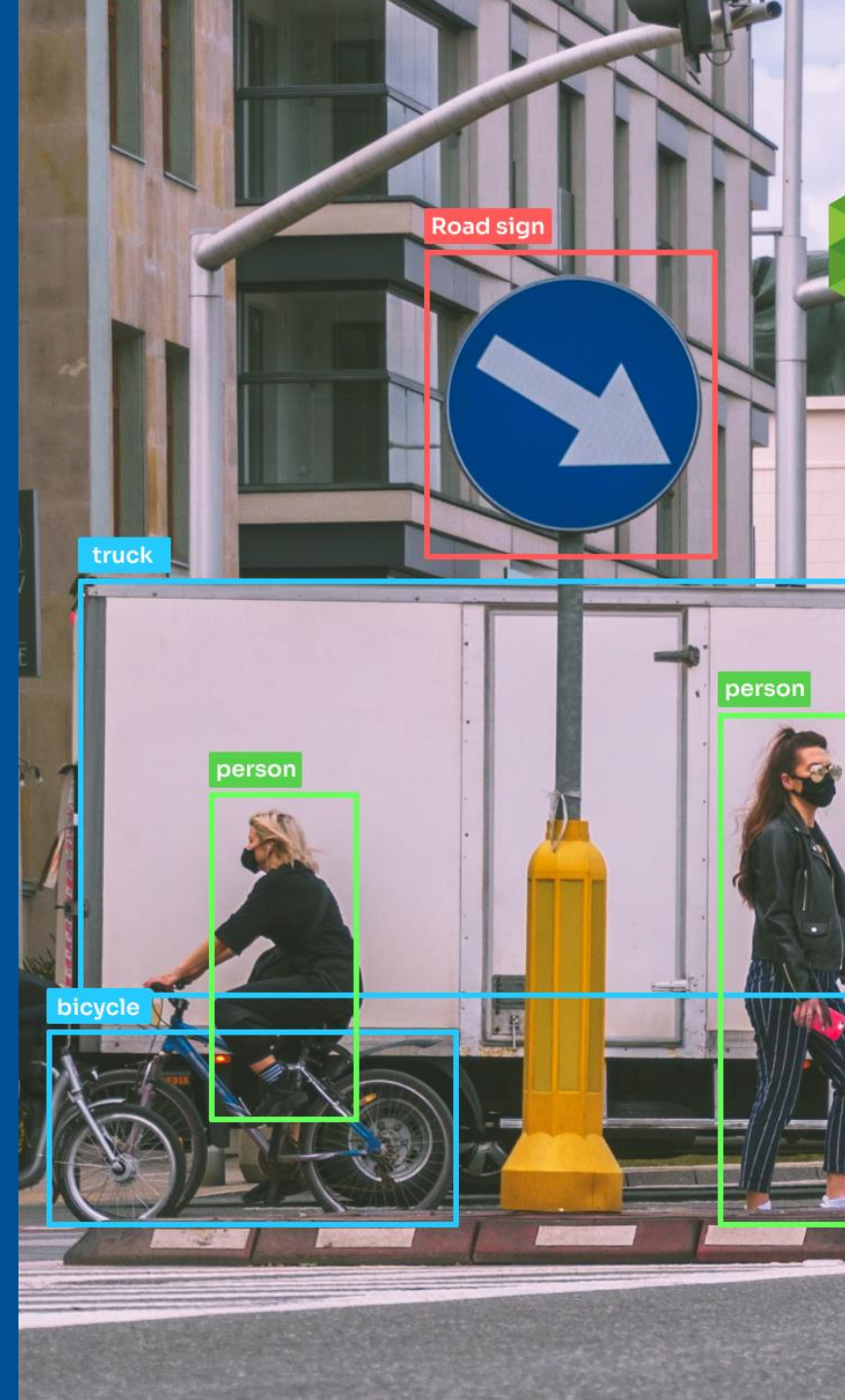
> "Warning: the vehicle is turning left!"

WHEEL ENCODER

> "The wheels are turning; forward motion active."

Each sensor provides only a single, isolated piece of the puzzle. Only the computer puts them together to reveal the whole picture.

AI for Image & Obstacle Recognition

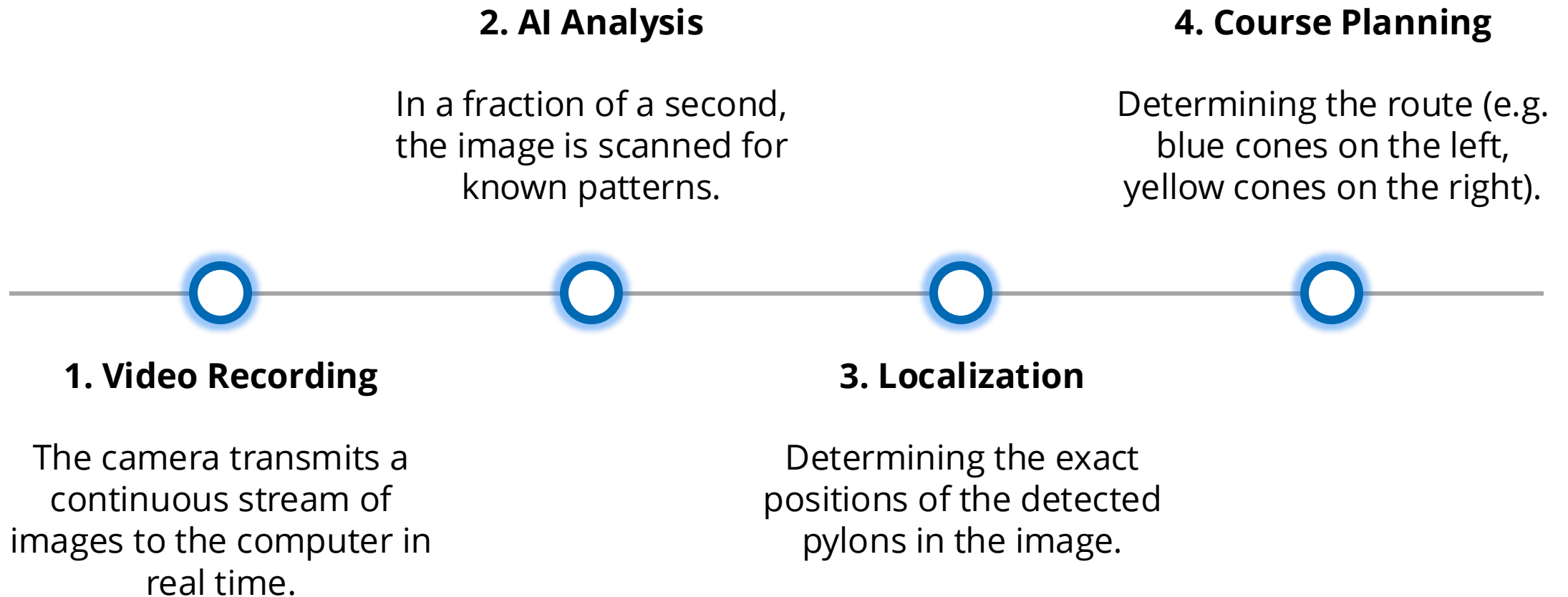


| The Illusion of Sight

- Cameras do not “see” traffic cones; they merely provide a grid of millions of color pixels.
- To a computer, an image is primarily a gigantic matrix of abstract numerical values.
- Artificial intelligence converts these pixels into usable information and bridges the gap to a true understanding of the environment.



| The Pylon Detection Pipeline



| The Evolution of Computer Vision

Traditional Image Processing

In the past, rigid rules were programmed: "Look for pixels that are orange and form a triangle."

The problem: It works well in the lab, but fails immediately in the real world when lighting conditions change.

Deep Learning (Neural Networks)

Instead of writing rules, the AI learns independently from thousands of training images.

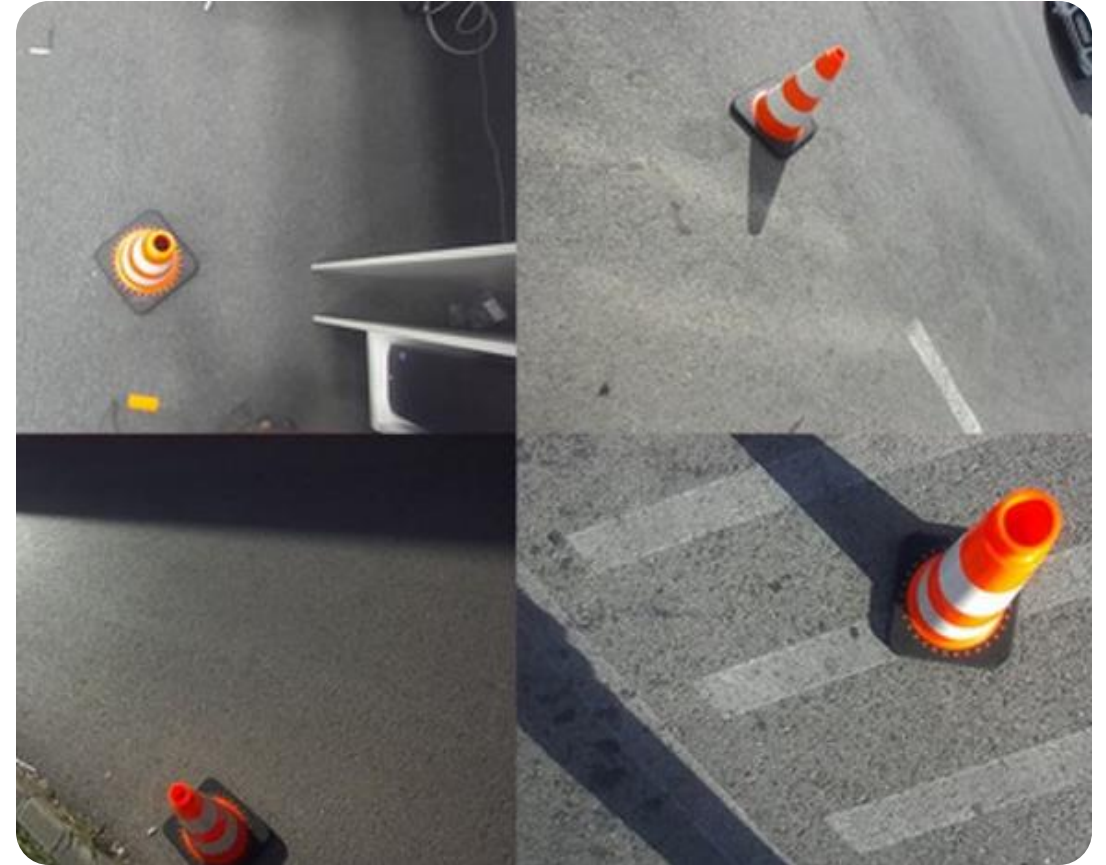
The advantage: The systems understand the abstract patterns and edges that define an object and perform reliably in practice.

| Object Recognition & Classification

The AI's goal is divided into two parts

First, the AI must locate the target object in the image stream and draw a precise frame around it—the so-called bounding box.

At the same time, it must classify the object it has found and calculate the level of certainty it has (e.g., “Pylon with 98% probability”).



| Two Phases of Artificial Intelligence



1. Training

The intensive learning process takes place in advance on massive server farms and often lasts for days. Here, the model learns from the training data.



2. Inference

The application of the trained model. While the car is in motion, new, unknown images are analyzed in real time, in milliseconds.



Evaluation

We continuously measure accuracy. The goal is to minimize the number of objects that are missed (false negatives) and incorrectly identified (false positives).

| Real-World Challenges



Lighting Conditions & Shadows

Pylons look completely different at dusk, in direct sunlight, or under harsh indoor lighting. Shadows can also drastically distort shapes.



Obstacles

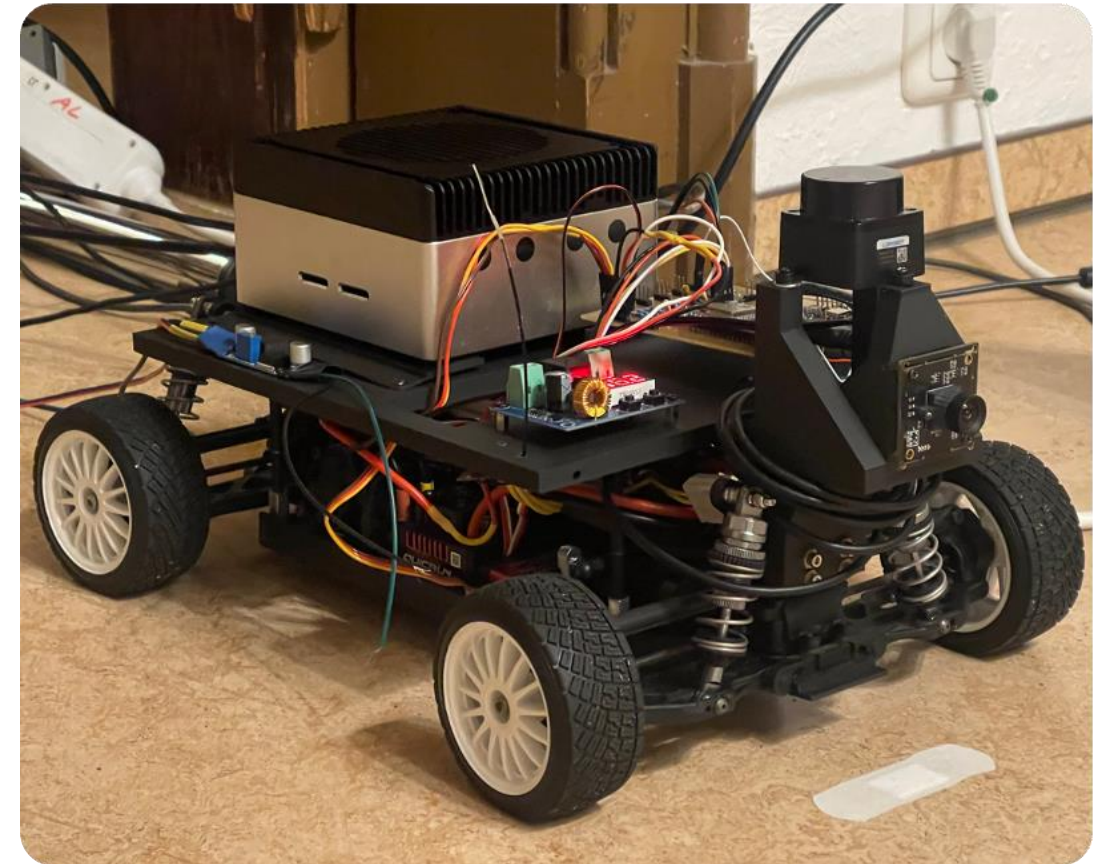
If one pylon is half-hidden behind another, the AI suddenly loses 50% of its visual identification cues.



Distance & Motion Blur

Objects 10 meters away consist of nothing more than tiny clusters of pixels. Fast turns cause the camera image to blur.

| Transfer to our RC car



Obstacle and Person Detection



| The Dynamic Environment

Beyond the Pylons

Pylons are easy to deal with: they stand still. The real world, however, is dynamic and completely unpredictable.

In addition to detecting people and vehicles, free-space detection is essential: the car must be able to precisely define the safe, drivable space.



| Recognition vs. Understanding

Pure Recognition

The camera AI reports: "That's a pedestrian over there."

The problem: This is a great first step in classification. However, it does not yet provide the autonomous vehicle with actionable information for a specific maneuver.

True Understanding

The system must also assess: "Where is the pedestrian physically located, and what will they do next?"

The goal: Only the combination of precise spatial coordinates and a prediction of behavior makes autonomous driving safe.

| Critical Questions



Position & Distance

Where exactly is the obstacle located in space, and how far is it from your vehicle? (We need coordinates, not just pixels.)



Movement & Dynamics

Is the object moving? If so, how fast and in which direction? (Is it a parked car or a running child?)



Trajectory

What path is the object expected to take in the next few seconds?

| Safety Responses & Escalation Levels



1. Prevention

In non-critical situations, a proactive adjustment is sufficient: The vehicle eases off the accelerator, reduces speed, and gently increases the safety distance.



2. Evasion

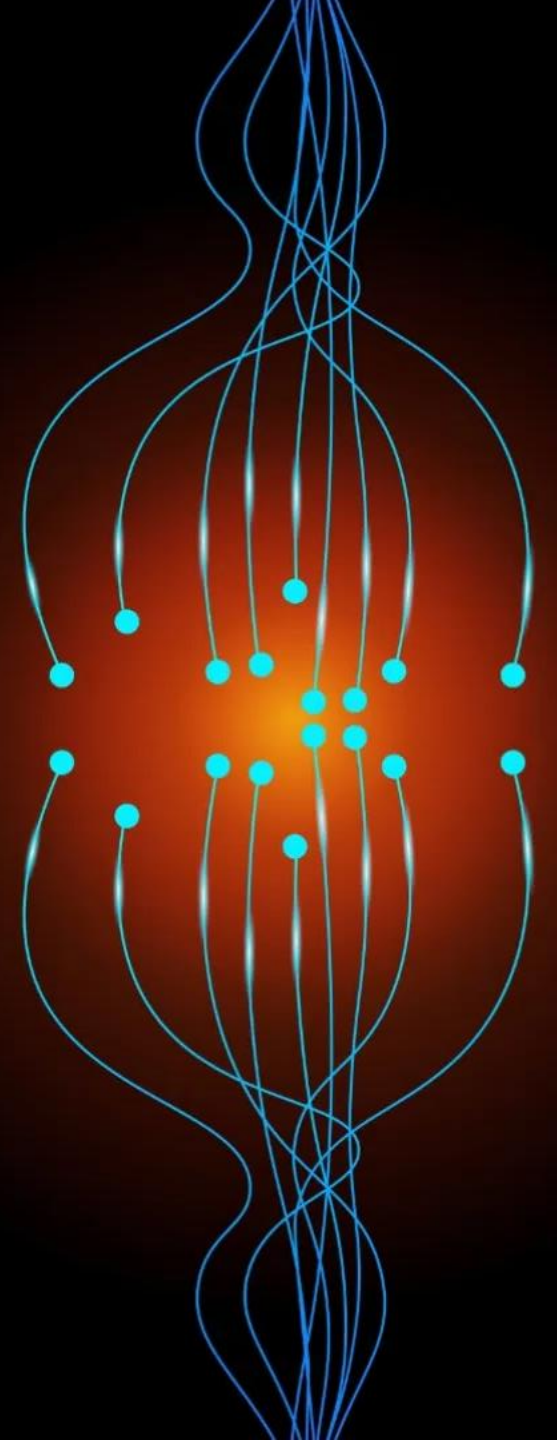
If the situation becomes tighter, the system calculates in milliseconds whether there is still enough space for a safe steering maneuver into the previously detected free space.



3. Emergency Braking

The ultimate last resort: If the braking distance becomes too short and there is no room to evade, the wheels are locked to minimize the impact energy.

Sensor Fusion



| The Concept of Sensor Fusion

Putting the Puzzle Together

No single sensor has a complete picture of reality. The camera recognizes categories, while the radar measures distances with precision.

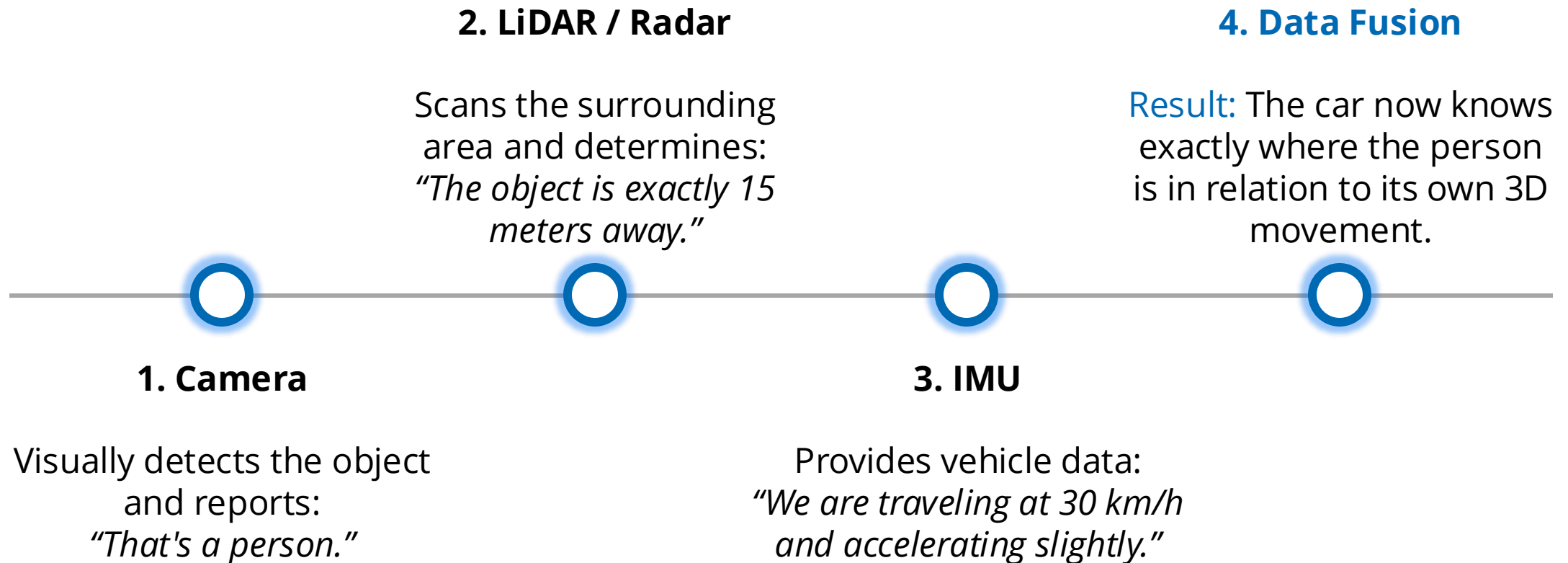
The basic idea behind sensor fusion is to combine these isolated data streams to generate a single, highly reliable representation of the environment.

1 Point Cloud 2D Projection



2 2D Object Detection

| Sensor Fusion in Practice



| Why all this computational effort?



Different Types of Errors & Failures

Every sensor has its weaknesses. If the camera fails due to the sun being low in the sky, the car would essentially be driving blind without sensor fusion.



Sensors Complement Each Other

Cameras provide the “what” (colors, object classes). LiDAR and radar provide the “where” (depth, speed). Together, they provide a complete picture.



Redundancy for Safety

If two independent sensors detect an obstacle, the probability of error is extremely low. This is crucial for regulatory approval.

| The Magic of Mathematics



State Estimation

Probabilistic Principle: The system never assumes absolute certainties, but instead constantly works with probabilities (where is the object likely to be?).



Kalman Filter

Operates in a continuous two-step cycle: It first makes a prediction and then, in the second step, compares this prediction with the actual measurement.



EKF

The Extended Kalman Filter is the enhanced version designed for practical applications. It was specifically developed to calculate nonlinear, complex movements (e.g., curves).

Localization

Where is the system located?

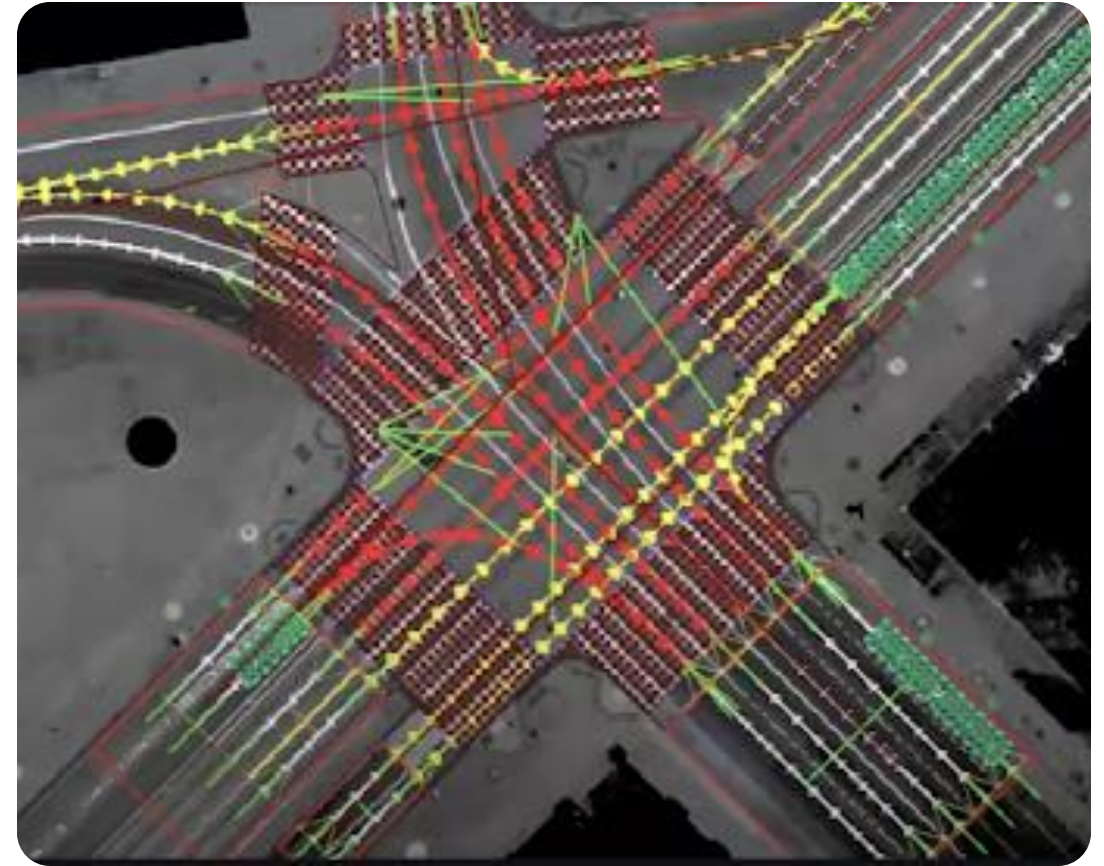


| Localization: Where am I?

The Missing Piece of the Puzzle

Thanks to object recognition and sensor fusion, the system now knows where obstacles are. But in order to start moving, one fundamental piece of information is missing.

Every autonomous system absolutely requires a reliable estimate of its own position and orientation (direction of view) in space.



| Sensors for Position Determination



GPS or GNSS

Provides our absolute position on the world map (latitude and longitude).



Wheel Encoders & IMU (Internal Sensors)

Track the vehicle's own movement and acceleration from within the vehicle.



Camera & LiDAR (Landmarks)

Scan the surroundings for distinctive, fixed points in the environment (buildings, traffic signs, or pylons).

| The Limitations of Odometry

Why don't we simply rely on measuring the distance we've traveled based on the number of wheel revolutions?

The theory: 10 wheel revolutions = exactly one meter traveled.

In practice: If the road is wet, covered in sand, or if a wheel is off the ground, it spins freely.

The result: The system “thinks” it has traveled, but in reality, it's still practically standing still.

The solution: We must combine external landmarks with various sensors to correct errors.



| Navigation Terminology



Positioning

Absolute vs. Relative: An absolute position is an exact GPS coordinate. A relative position, on the other hand, is local: "I am 5 meters straight ahead and 2 meters to the right of my starting point."



Drift

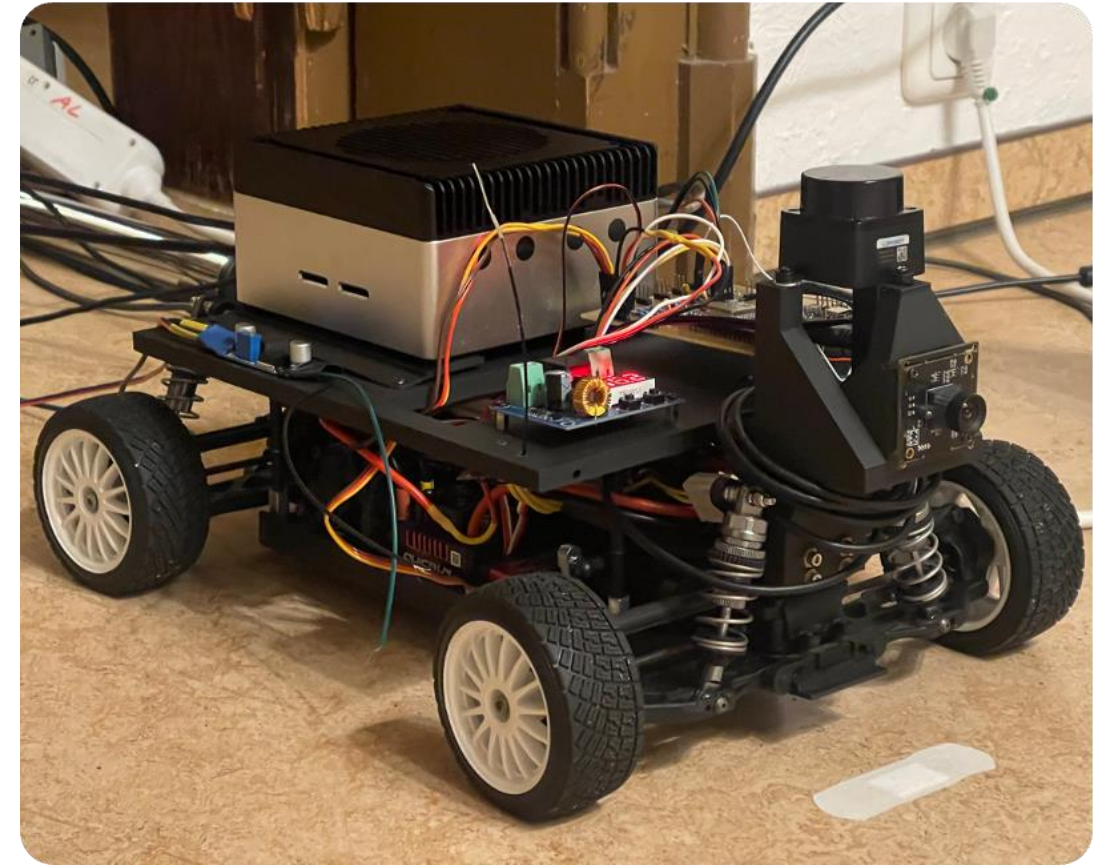
Error Accumulation: A spinning wheel causes a minimal measurement error. This error does not disappear but accumulates meter by meter until the system drifts completely out of sync with reality.



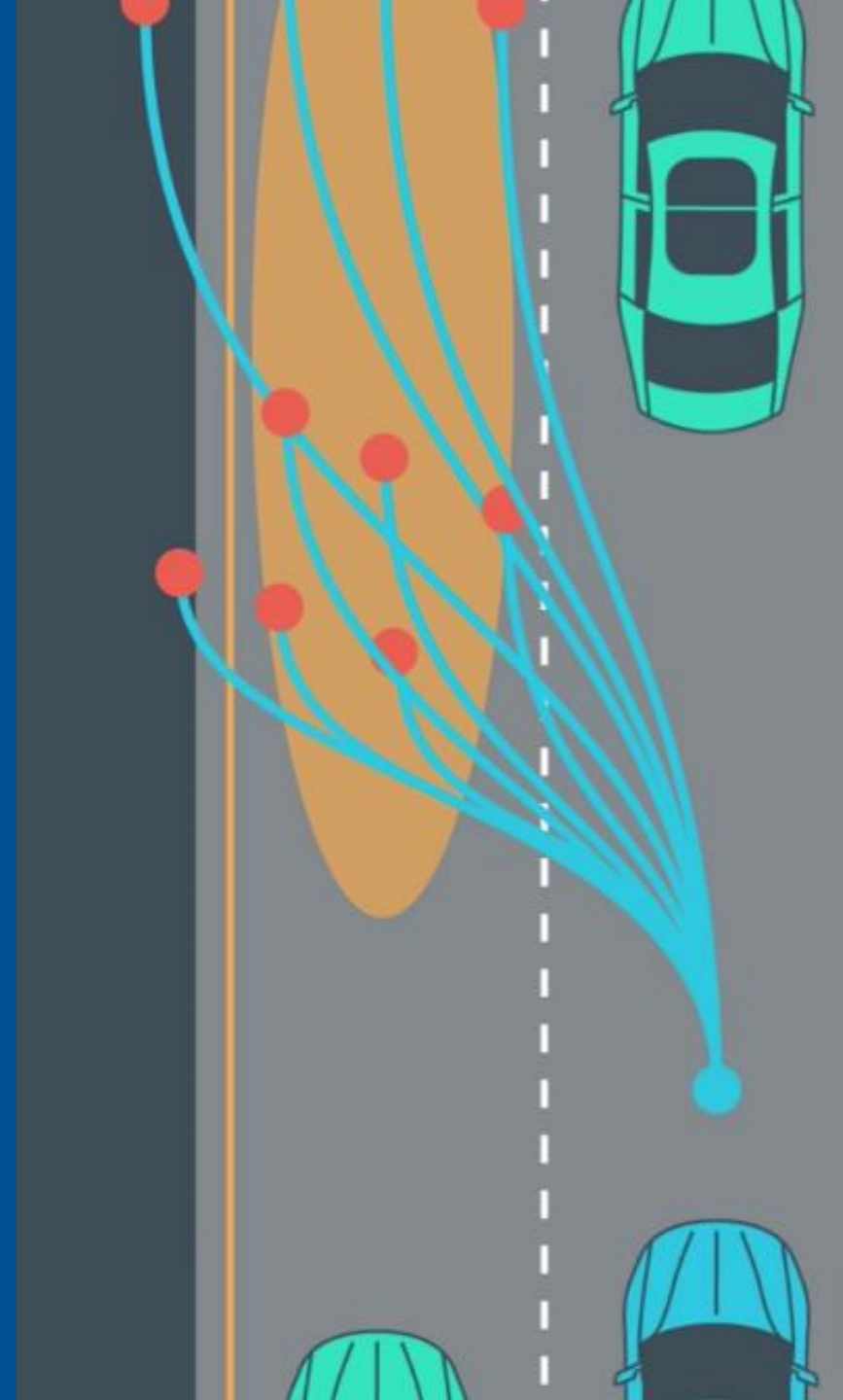
SLAM Correction

To counteract this fatal drift, SLAM continuously compares the real-time map with actual landmarks and mathematically "pulls" the vehicle back on course.

| Adapting to Our RC Car



Path and Trajectory Planning

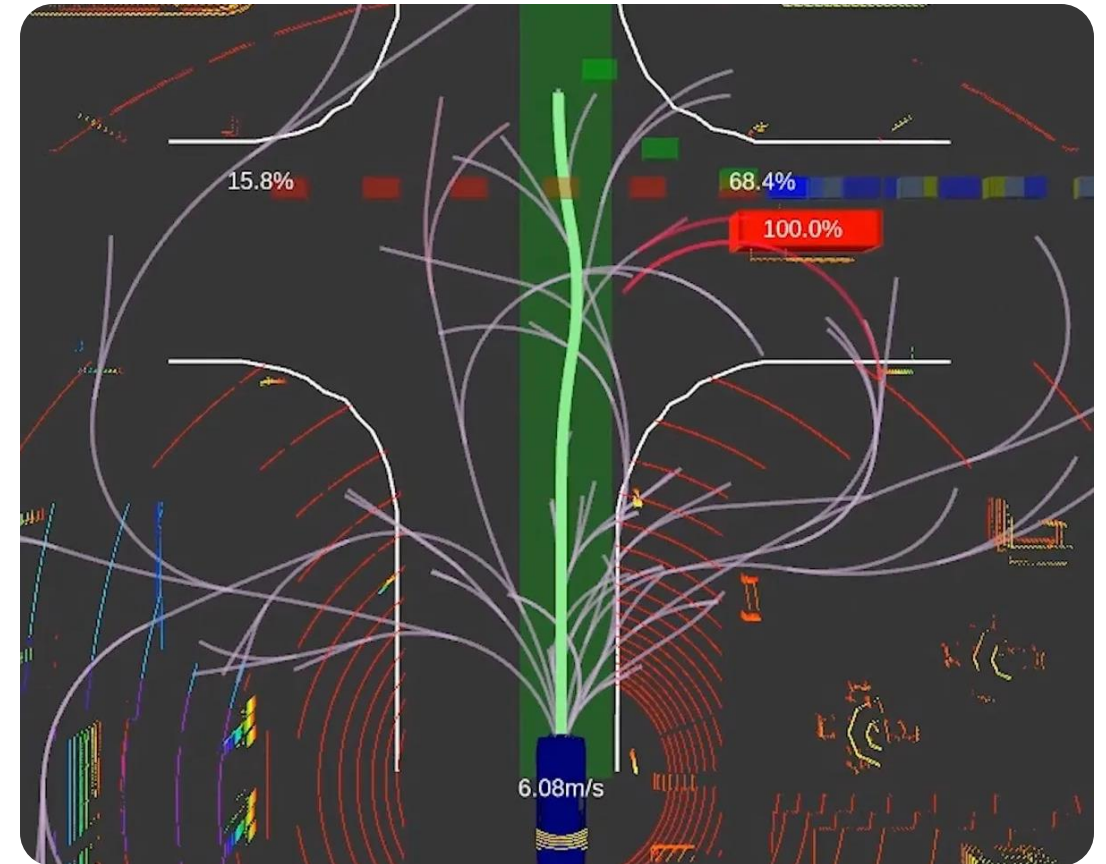


| The logical next fundamental question

Perception is complete. Our vehicle now knows exactly:

- Location: "I am right here."
- Environment awareness: "This is what's around me."

"How do I get to my destination safely and efficiently?"



| Classic Search Algorithms



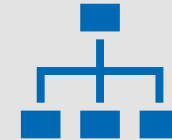
Dijkstra

Systematically searches all possible paths. Very thorough, but often computationally intensive, since it spreads evenly in all directions.



A*

Performs a goal-oriented search and uses heuristics for estimation. As a result, it finds the optimal path significantly faster than undirected searches.



RRT

Rapidly-exploring Random Trees. Navigates through complex spaces like a network of branches. Ideal for high-dimensional and dynamic environments.

| The Two Levels of Planning

Global Path

Long-term route planning from start to destination.

- Similar to a standard navigation system.
- E.g., “Drive from Munich to Berlin.”
- Focuses on the road network.

Local Path

Short-term planning for the immediate surroundings (e.g., the next 50 meters).

- Real-time collision avoidance.
- Response to dynamic obstacles (e.g., pedestrians).
- Lightning-fast recalculation in response to changes.

i Pathfinding is often brilliant classical mathematics—not necessarily AI!

| Definition of Terms



Path Planning

Addresses the question of geometry.

Where should I generally go? Path planning essentially draws a continuous line on the ground that avoids obstacles but ignores physical dynamics.



Trajectory Planning

Addresses the question of kinematics.

How precisely should I follow this path? It necessarily adds a time and velocity stamp (steering angle, acceleration) to the geometric line.

| Possible Optimization Goals



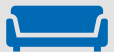
Time & Distance

Finding the shortest or fastest route, depending on traffic conditions and the route profile.



Energy Efficiency

Minimizing energy consumption for maximum range through anticipatory acceleration.



Ride Comfort

Low cornering forces and smooth braking maneuvers to prevent passengers from feeling nauseous.



Safety & Physics

Maintaining a sufficient distance from obstacles and strictly adhering to the vehicle's limits (loss of traction).

| The Race Track Example

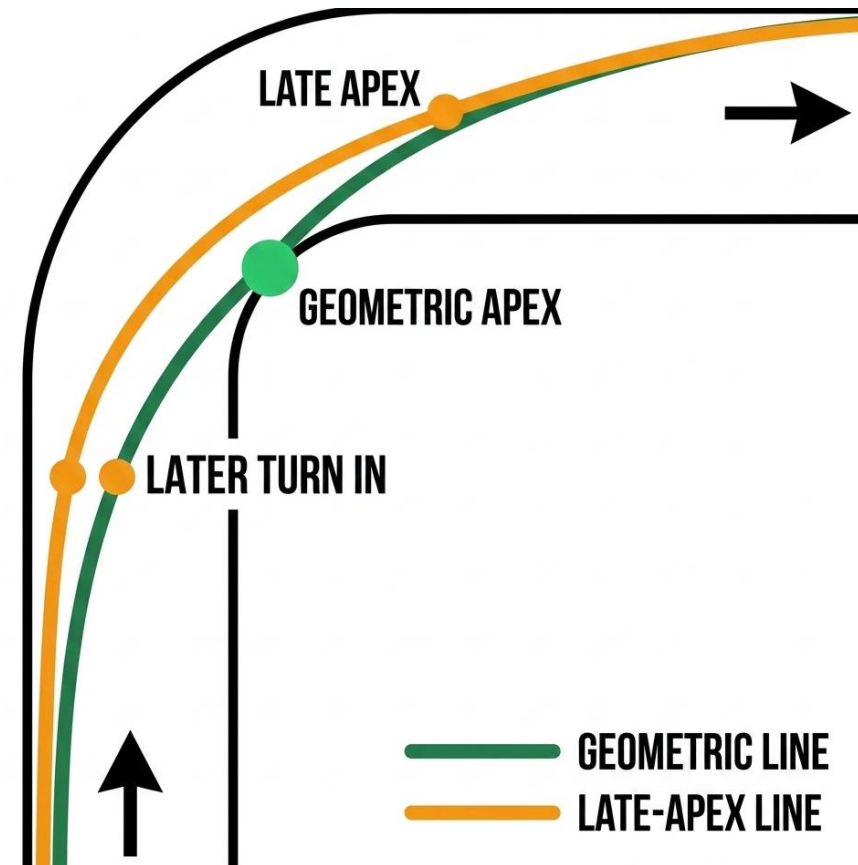
Different optimization goals lead to completely different trajectories, even though the path lies within the same corridor.

1. The Geometric Center

Maintains exactly the same distance from both edges. This is the safest route with the maximum tolerance range.

2. The Ideal Line

A race car driver cuts into turns, brakes late, and uses the entire width of the track. This is the fastest line and the one that offers the best driving dynamics.



| The Continuous Control Loop

The controller constantly compares the desired path with the actual path and asks itself the following questions in a fraction of a second:



Lateral Deviation

Am I currently to the left or right of my desired line?



Steering Angle

How much do I need to turn the wheel right now to get back exactly on the line?



Speed

How fast can I go into the next turn without losing control?

| Vehicle Control Methods



PID Controller

The classic. It has a simple design but often responds somewhat jerkily, since it primarily reacts only to the currently measured error.



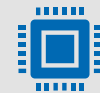
Pure Pursuit

Like a carrot in front of a donkey. It selects a point on the path some distance away and steadily steers directly toward it.



Stanley Controller

Won the 2005 DARPA Grand Challenge. Calculates errors extremely smoothly and is excellent for high speeds.



Model Predictive Control (MPC)

The modern gold standard. Looks ahead and simulates the vehicle's physical behavior based on current steering inputs.

State of the Art



| The Architecture War

Modular AI

The classic pipeline:

Sensors → Perception → Sensor Fusion → Planning → Control.

Advantage: 100% interpretability. Engineers know exactly which module made a calculation error (verifiability for regulatory authorities).

Disadvantage: Errors accumulate. Millions of lines of code are required to enforce strict rules.

End-to-End Learning

Software 2.0 (Behavioral Cloning):

Raw video stream in → direct steering & braking commands out.

Advantage: Scales infinitely. Driving is incredibly smooth and “instinctive.”

Disadvantage: An absolute black box. No one knows in detail why the neural network made that decision in that millisecond.

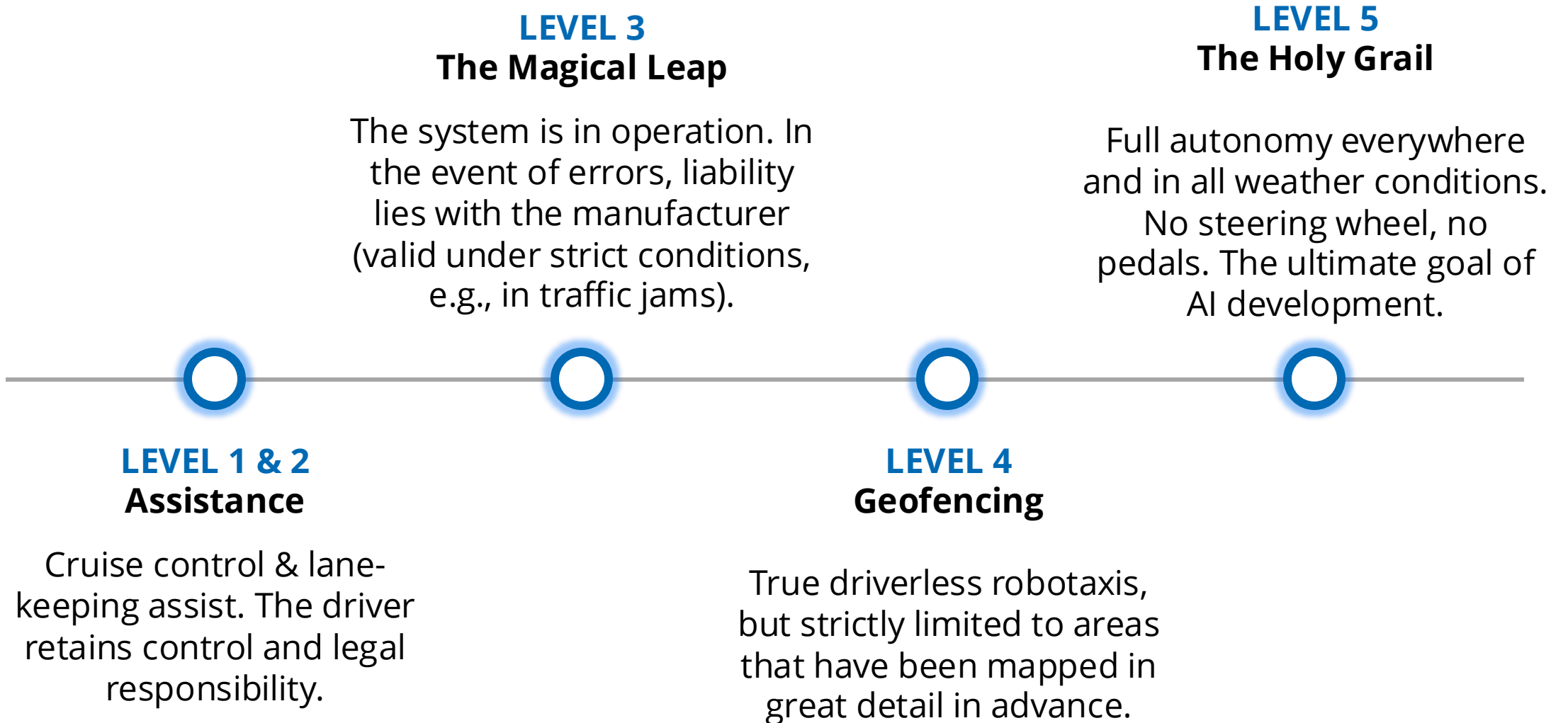
| Vision-Language-Action (VLA)

“I see a dog off-leash. Animals are unpredictable, so I’m slowing down as a precaution.”

— AI with real-time logical reasoning capabilities

Today, hybrid systems are breaking out of the “black box” by directly integrating language models. The system is developing a form of “common sense.”

| The Evolution of Autonomous Driving



| Milestones & Players 2026



WAYMO

Waymo

The undisputed market leader. Highly complex hybrid systems with LiDAR. Over 200 million miles and nearly 4,000 driverless taxis (under regulatory oversight).



| Milestones & Players 2026



Tesla

Going “completely against the grain.” No LiDAR, purely camera-based end-to-end. Start of production for the long-awaited “Cybercab” (without a steering wheel or pedals).



| Milestones & Players 2026



Baidu

Dominant in 15 major metropolitan areas with the "Apollo Go" fleet. The harsh reality became apparent with the "Wuhan Paradox" (nearly 100 vehicles stranded).

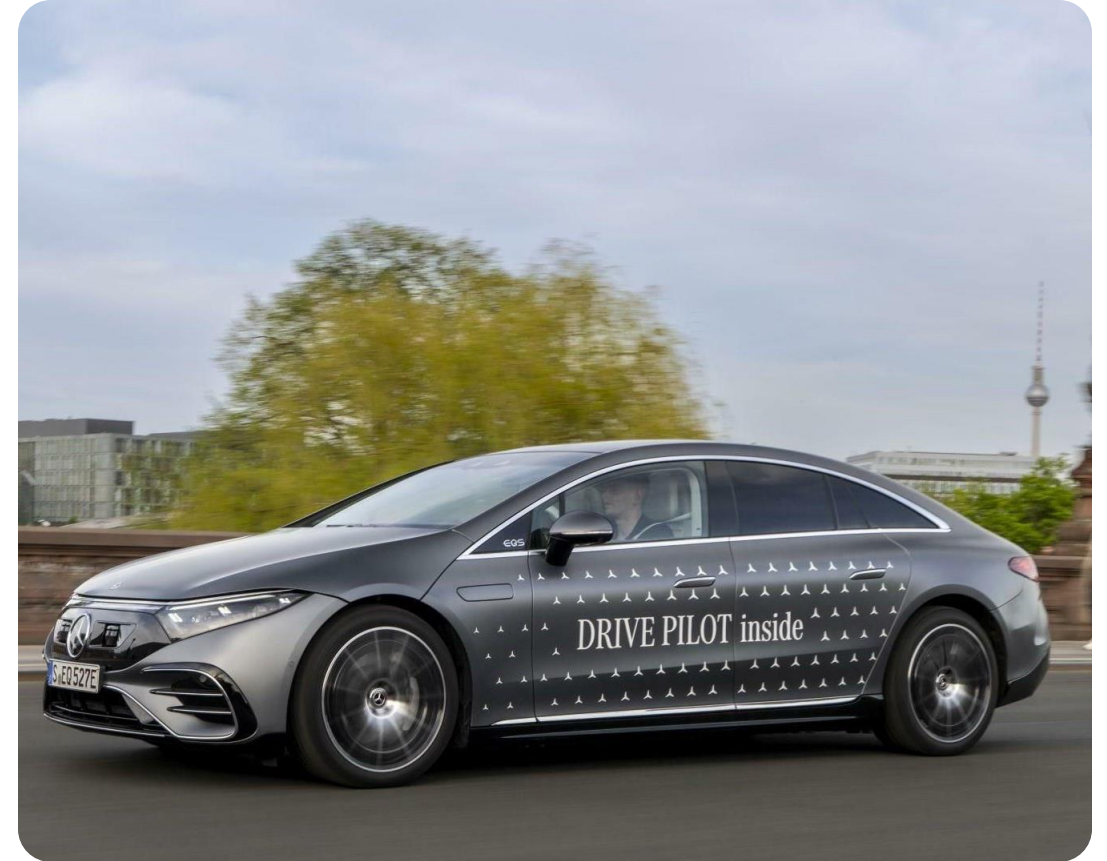


| Milestones & Players 2026



Mercedes-Benz

Focus on private customers rather than robotaxis. When activated, the “Drive Pilot” (Level 3) is the first in the world to assume full legal liability for the driver.



| The Long-Tail Problem

The Taxonomy of the Unpredictable

Self-driving cars excel at routine tasks. However, they falter when faced with edge cases and extreme corner cases.

⚡ A corner case (heavy rain + winding road + heavy traffic + sudden braking) is unlikely, but it overwhelms the system massively.

It is statistically impossible to accumulate enough real-world test kilometers to capture every absurd exception case on video for the SOTIF standard.



| Training in the Matrix

How is research addressing the data deficit?
Through world models.



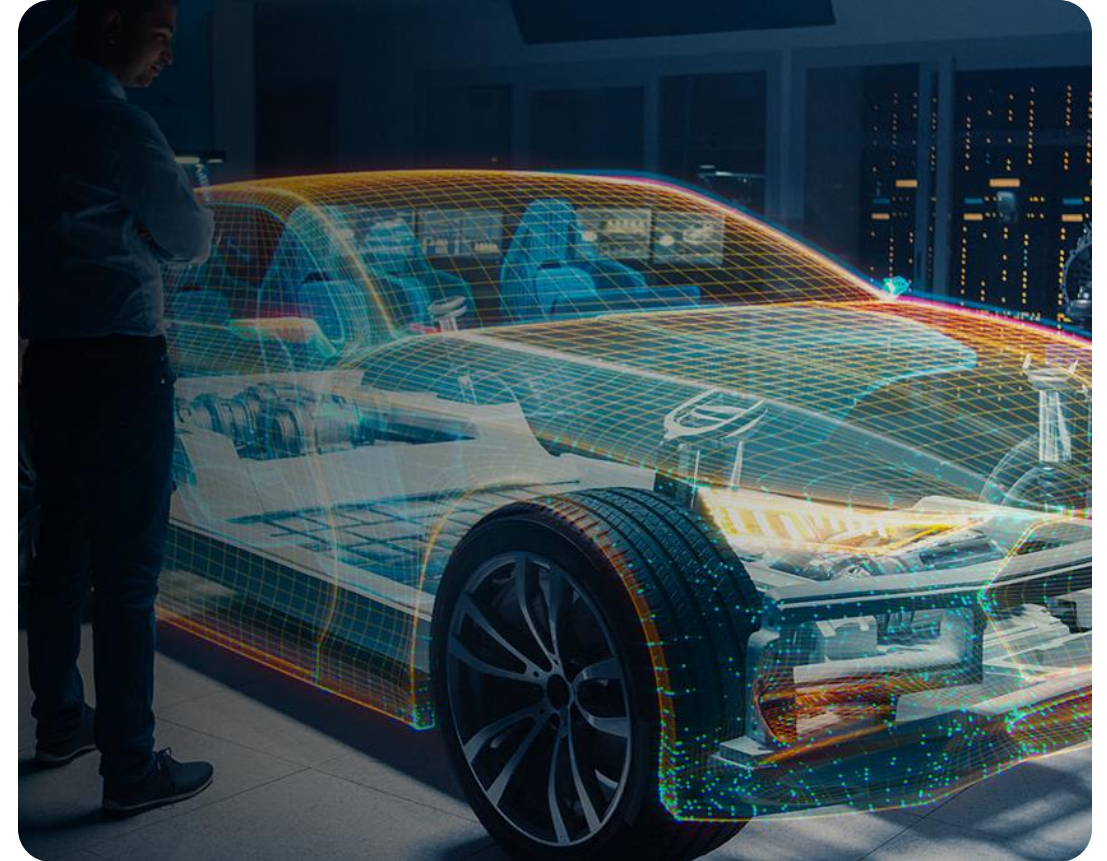
Virtual reality: AIs simulate photorealistic worlds (digital twins).



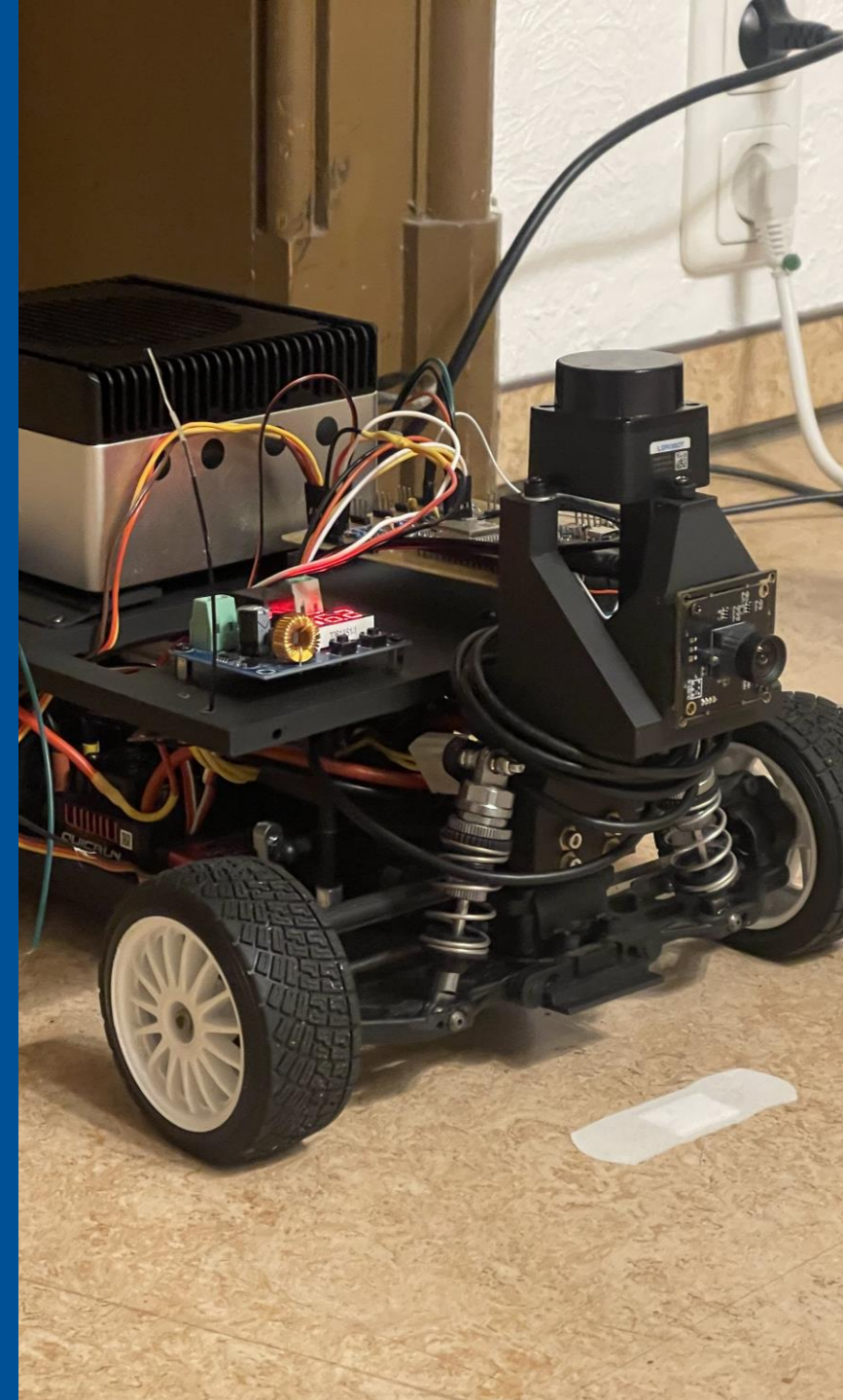
Safe training: The car runs through millions of alternative future scenarios without risking accidents.



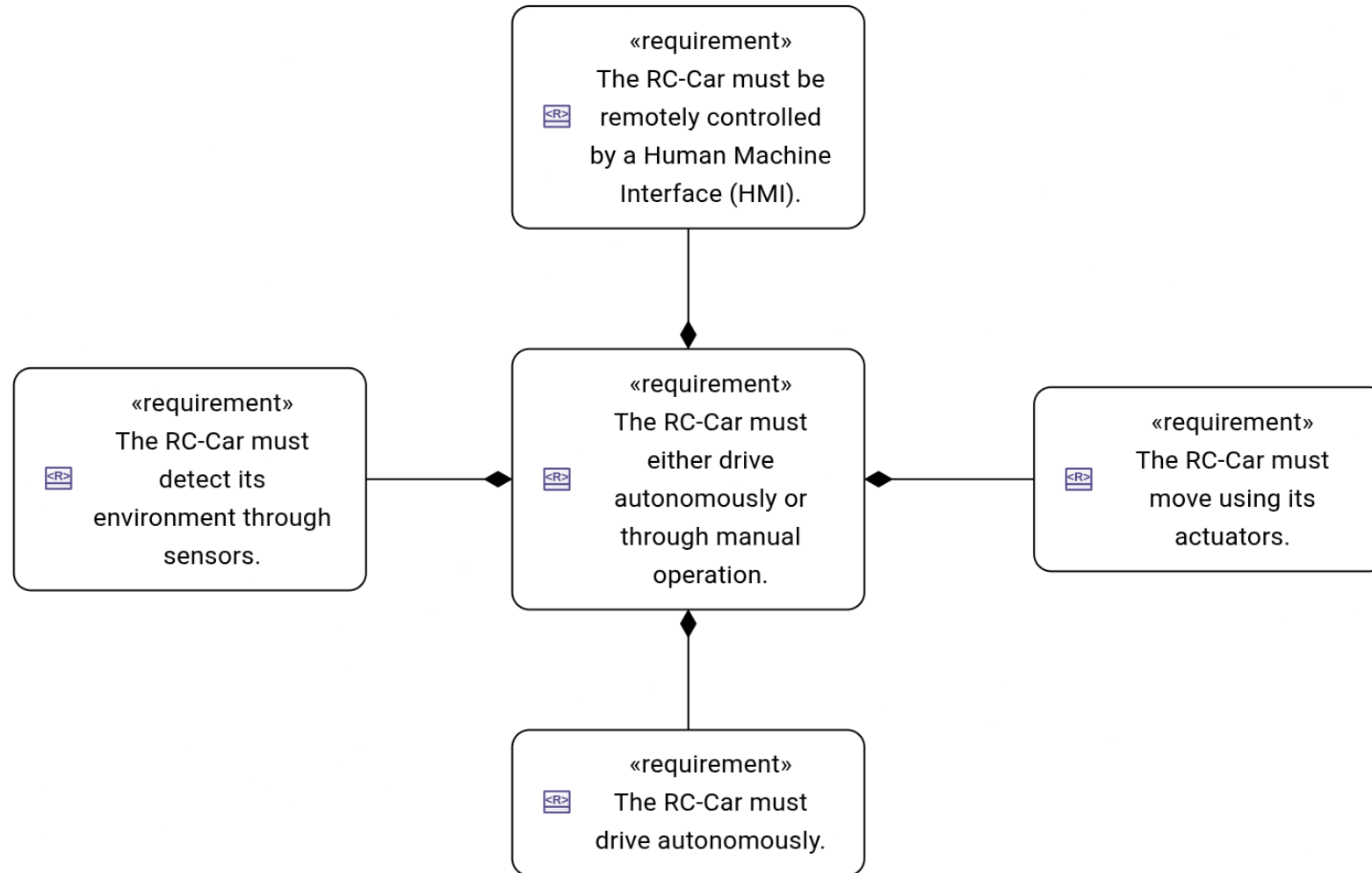
Intuition: Coupled with language-based logic models (MLLMs), AI is approaching human intuition.



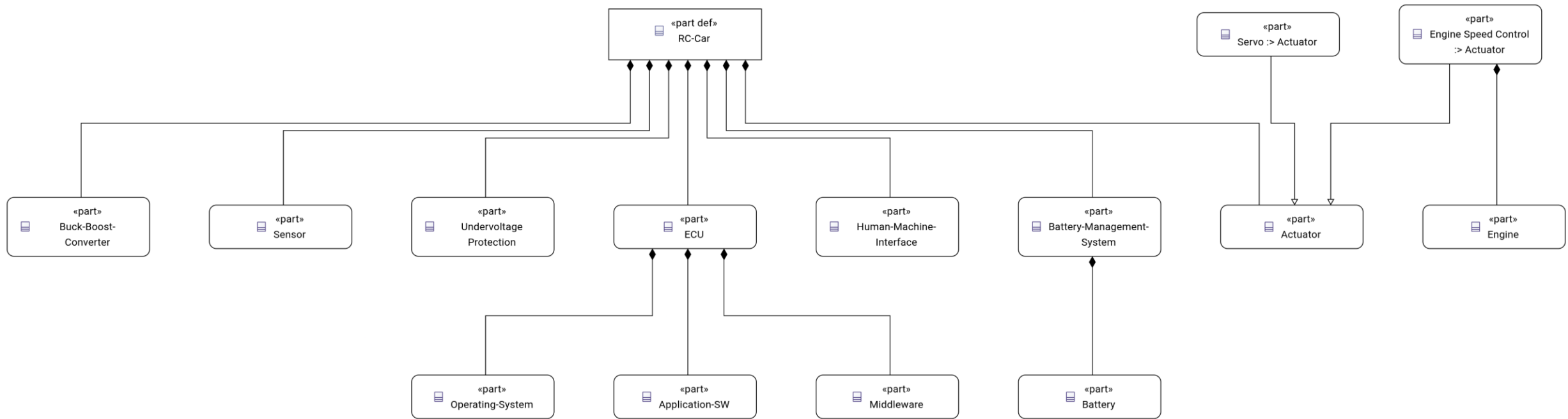
Our RC Car



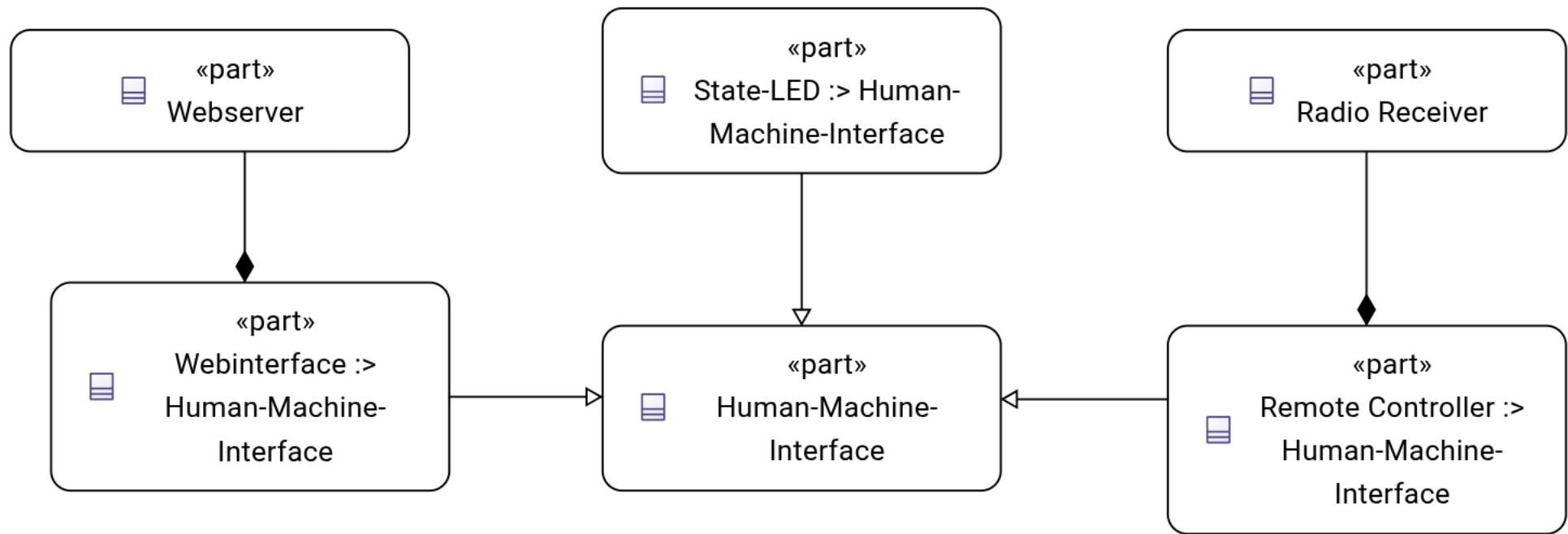
| General Requirements



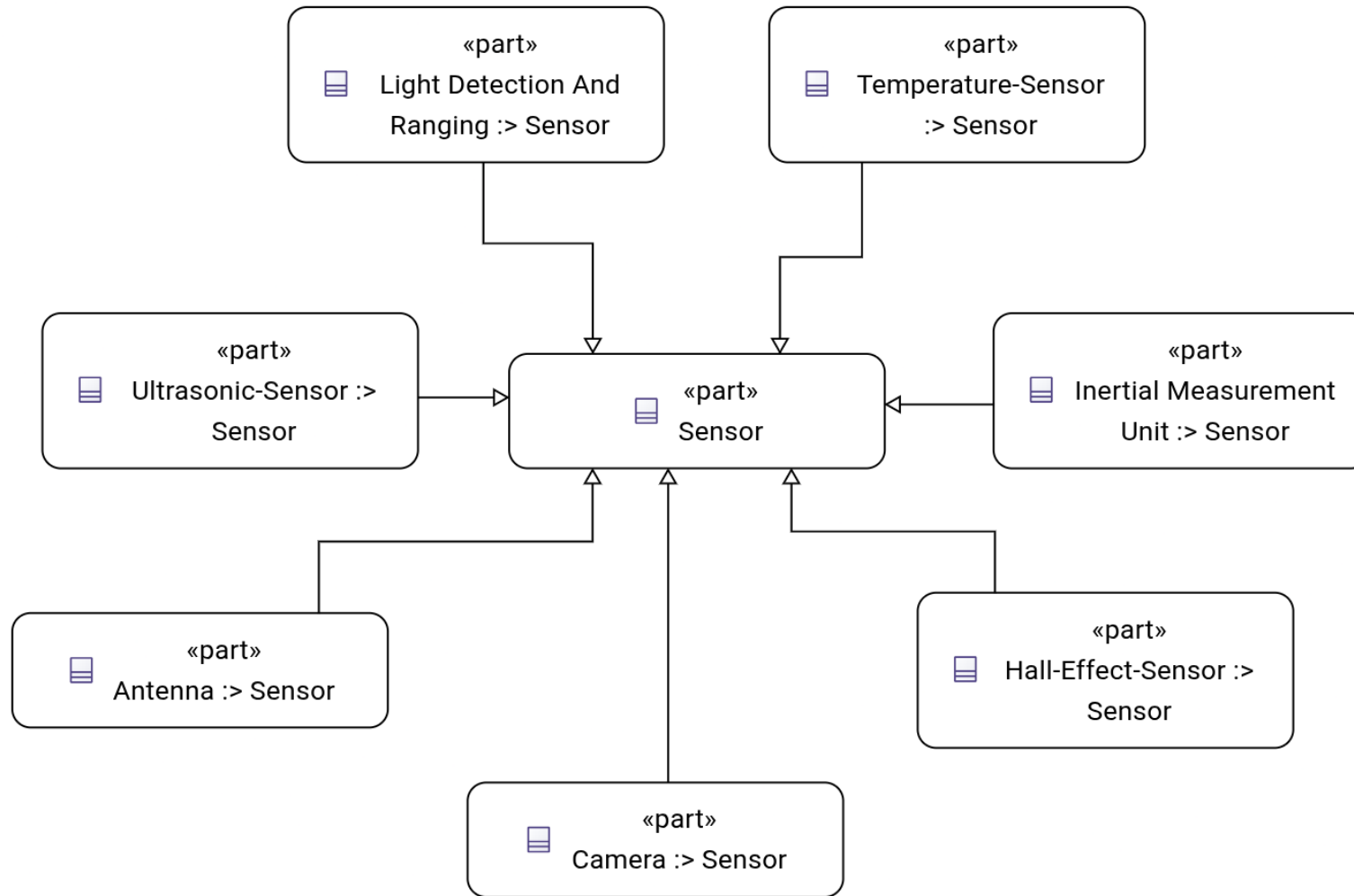
| System Domain



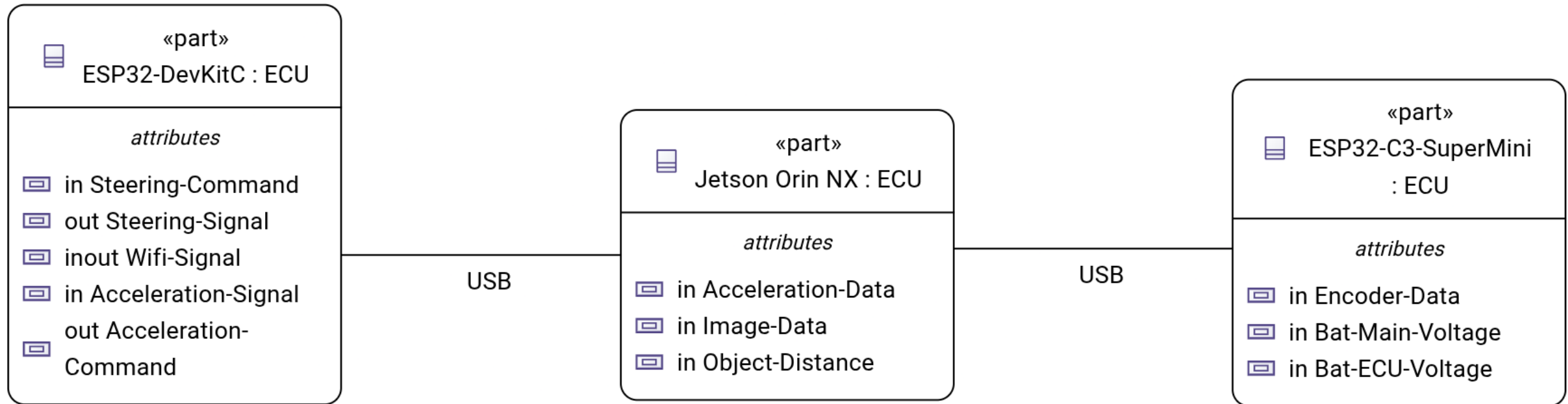
| HMI Domain



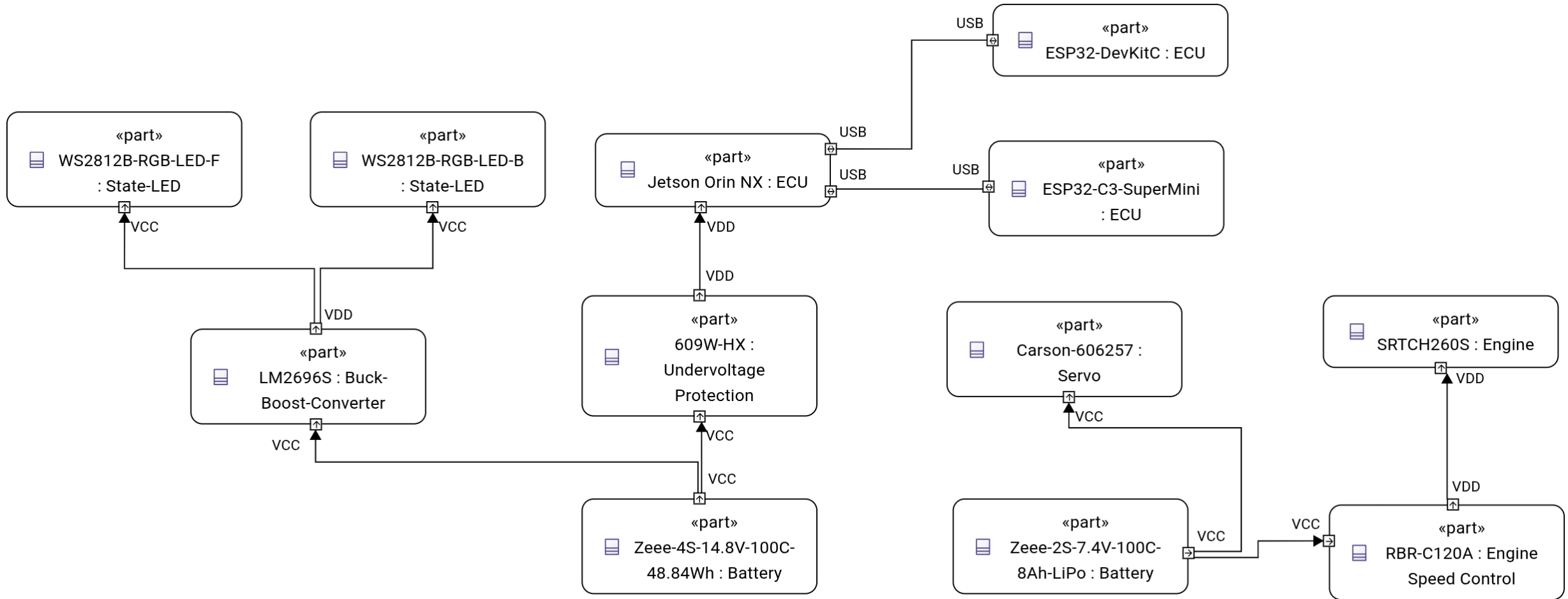
| Sensor Domain



| Hardware Architecture



| Power Architecture



| Software Architecture

| Sequence Diagrams

| Results

| Achieved requirements

| Failed requirements



Conclusion





Thank you



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